Solution

A1. From thermodynamics we know that $v = \sqrt{3kT/m}$, where m is the particle's mass and k is the Boltzmann constant. The characteristic jump time between inter-nodes is inversely proportional to the transition probability times the particle's speed, and the diffusion coefficient is inversely proportional to the jump time; therefore

$$\boxed{v = \sqrt{\frac{3kT}{m}}, \qquad D(T) \propto v \exp(-E_g/kT) \propto \sqrt{T}\exp(-E_g/kT)}$$

A2. First, let us find the rms particle displacement along one of the axes (call it x). The motion along it happens as in the one considered in the preparatory lectures, so the mean square displacement is

$$\overline{x^2} = na^2,$$

where n is the total number of jumps. The number of jumps can be estimated as $n = t/\tau$. With $D = a^2/2\tau$ we obtain

$$\overline{x^2} = 2Dt.$$

Finally, because the motions along different axes can be considered independent, $\overline{y^2} = \overline{z^2} = 2Dt$. By the Pythagorean theorem,

$$\boxed{\Delta r_{sq} = \sqrt{\overline{x^2} + \overline{y^2} + \overline{z^2}} = \sqrt{6Dt}}$$

A3. Let us use the result of the previous task to find the diffusion coefficient. Because $\Delta r_{sq}^2 = 6Dt$, $D = \Delta r_{sq}^2/6t$. Combining this with A1, $\Delta r_{sq}^2/(6t\sqrt{T}) \propto \exp(-E_g/kT)$. The easiest way now is to find the slope of the dependence of $Y = \ln\left(\Delta r_{sq}^2 t^{-1} T^{-1/2}\right)$ on $X = 1/T$.

How the accuracy is guaranteed. First, it is good to have $\Delta r_{sq} \gg 1$. Second, $1/T$ changes by 0.9, and E_g is proportional to the slope. To guarantee $2\,\%$ accuracy the errors at the minimum and maximum temperatures must satisfy $\sigma_Y(0.1) + \sigma_Y(1.0) \le 0.02 \cdot 0.9 E_g$. Because $Y = \ln[\Delta r_{sq}^2 T^{-1/2}]$ (up to a constant), this means $2\varepsilon_r(0.1) + 2\varepsilon_r(1.0) \le 0.018 E_g$, i.e. it suffices that $\varepsilon_r = \sigma_r/\Delta r_{sq} \le 0.0045 E_g$ at every point. The data points:

t	10	10	10	10	10	10	10	10	10	10
T	1.000	1.111	1.250	1.429	1.667	2.000	2.500	3.333	5.000	10.00
N	10000	10000	10000	10000	10000	10000	10000	10000	10000	10000
Δr_{sq}	13.750	17.009	20.882	25.999	32.154	40.362	50.733	65.932	86.227	122.632
σ_r	0.053	0.066	0.081	0.102	0.127	0.156	0.198	0.257	0.336	0.477
X	1.000	0.900	0.800	0.700	0.6	0.5	0.4	0.3	0.2	0.1
Y	2.939	3.312	3.664	4.035	4.383	4.747	5.092	5.473	5.807	6.164

One can see that the required accuracy is indeed achieved. Plotting $Y(X)$ and finding its slope with `InteractiveBuffer.nb`:



	1	2
1	1.	2.939
2	0.9	3.312
3	0.8	3.664
4	0.7	4.035
5	0.6	4.383
6	0.5	4.747
7	0.4	5.092
8	0.3	5.473
9	0.2	5.807
10	0.1	6.164

The fit of $Y(X)$ in `InteractiveBuffer.nb`.

$$-E_g = -3.58 \pm 0.02 \quad \Rightarrow \quad E_g = 3.58 \pm 0.02$$

> Editor's note: the question paper numbers this task A2 a second time; the answer sheet and this solution call it A3.

B1. From the question paper we can conclude that the grain boundaries perpendicular to the diffusion flow do not affect the effective diffusion coefficient. Of the three families of boundaries (total relative volume δ), the two parallel to the diffusion direction carry $2\delta/3$, so

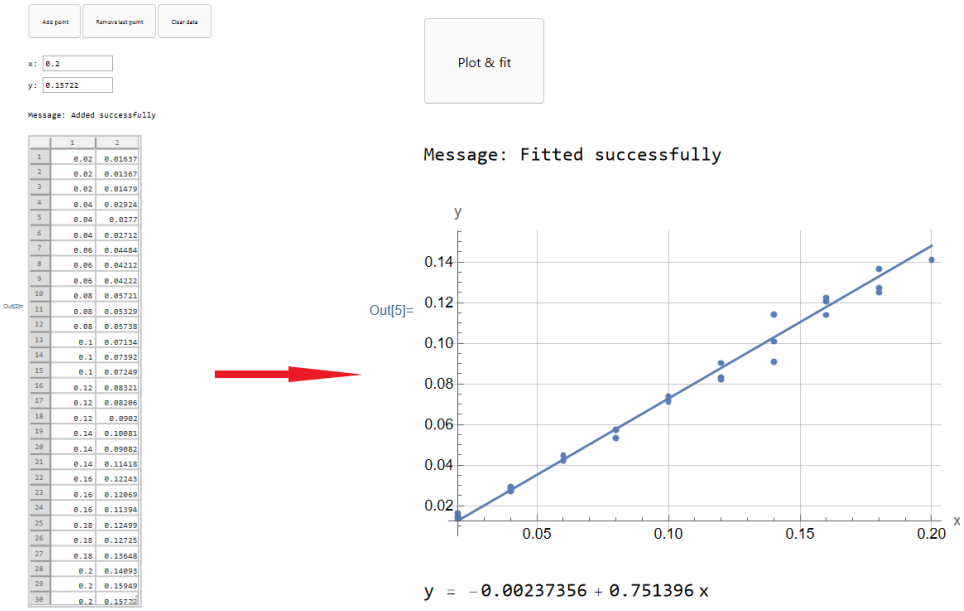
$$D_{eff} \approx \frac{2\delta D_{gb}}{3} \quad \Rightarrow \quad \alpha = \frac{2}{3}$$

B2. The simulation results are better for larger D_{gb}/D and smaller δ. It is also better to use the maximum simulation time, since type A diffusion appears on large time scales. So let us take $(t, D_{gb}/D, \delta) = (5, 5, 0.005)$ as the starting point. The plots in `B.nb` should ideally have the slope $k = -1/(4D_{eff}t)$, so in the program's units

$$D_{eff} = \frac{1}{4|k|t} = 1 + \alpha\delta(D_{gb}/D - 1).$$

With the coordinates $X = \delta(D_{gb}/D - 1)$ and $Y = \dfrac{1}{4|k|t} - 1$ this dependence is linear. The best way to make X change considerably is to increase δ. Lowering the *Width* slider in `B.nb` also helps against edge effects. The data points ($t = 5$, $D_{gb}/D = 5$):

δ	X	$\lvert k\rvert$ **(three runs)**			Y		
0.005	0.02	0.049194	0.049325	0.049271	0.01637	0.01367	0.01479
0.010	0.04	0.048579	0.048652	0.048679	0.02924	0.02770	0.02712
0.015	0.06	0.047854	0.047978	0.047974	0.04484	0.04212	0.04222
0.020	0.08	0.047294	0.047470	0.047286	0.05721	0.05329	0.05738
0.025	0.10	0.046670	0.046558	0.046616	0.07134	0.07392	0.07259
0.030	0.12	0.046159	0.046208	0.045863	0.08321	0.08206	0.09020
0.035	0.14	0.045421	0.045837	0.044876	0.10081	0.09082	0.11418
0.040	0.16	0.044546	0.044615	0.044885	0.12243	0.12069	0.11394
0.045	0.18	0.044445	0.044355	0.043995	0.12499	0.12725	0.13648
0.050	0.20	0.043823	0.043122	0.043207	0.14093	0.15949	0.15722



The fit of $Y(X)$ in `InteractiveBuffer.nb`.

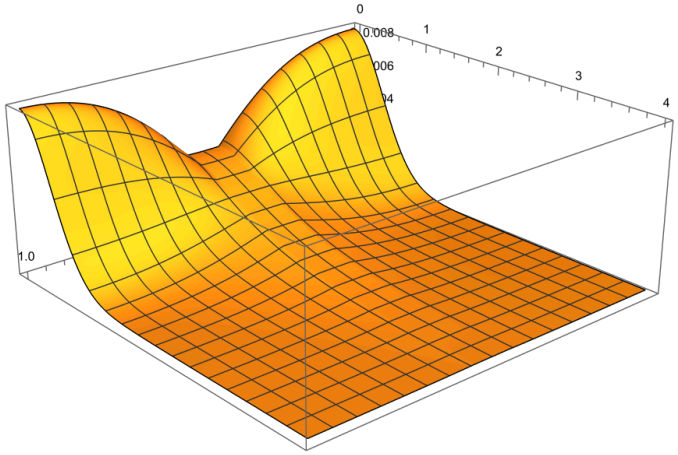
$$\alpha = 0.75 \pm 0.09$$

This coincides with the theoretical estimate of B1 within the margin of error.

C1. The only a priori constraint here is the dimensional one. Every factor on the right-hand side except a is dimensionless, so

$$\psi = -\frac{6}{5}$$

C2. As stated in the question paper, the scale of type B diffusion must be much smaller than a (i.e. $\sqrt{2[D_{gb}/D]t} \ll 1$) and at the same time much larger than δa (i.e. $\sqrt{2t} \gg \delta$). If we take, for example, the input parameters $(t, D_{gb}/D, \delta) = (0.100, 25, 0.10)$, we can obtain type B diffusion while varying any of the input parameters in a wide range. Whether the simulated diffusion was of type B can always be checked by plotting the concentration profile and comparing it with the one in the question paper. For example, the concentration map for $(0.100, 25, 0.10)$:



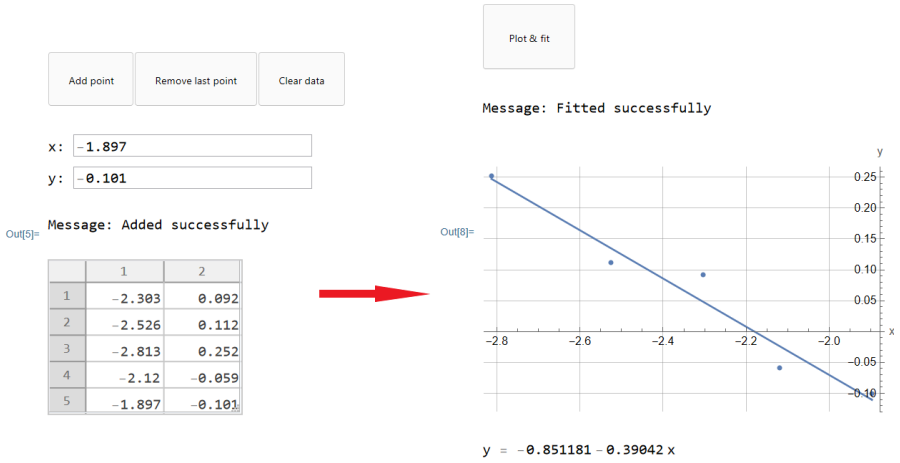
Concentration map for $(t, D_{gb}/D, \delta) = (0.100, 25, 0.10)$.

To find the slope $k = d\ln\bar{c}/dz^{6/5}$ we clip the linear regions of the plots generated in `C.nb`. ξ is obtained by simulating diffusion with different δ and plotting $\ln|k|(\ln\delta)$ in `InteractiveBuffer.nb`; η and ζ are found in the same way, varying D_{gb}/D and t. The data points (with $Z \equiv \delta^{-0.39}[D_{gb}/D]^{-0.68}t^{-0.44}$, used in C3):

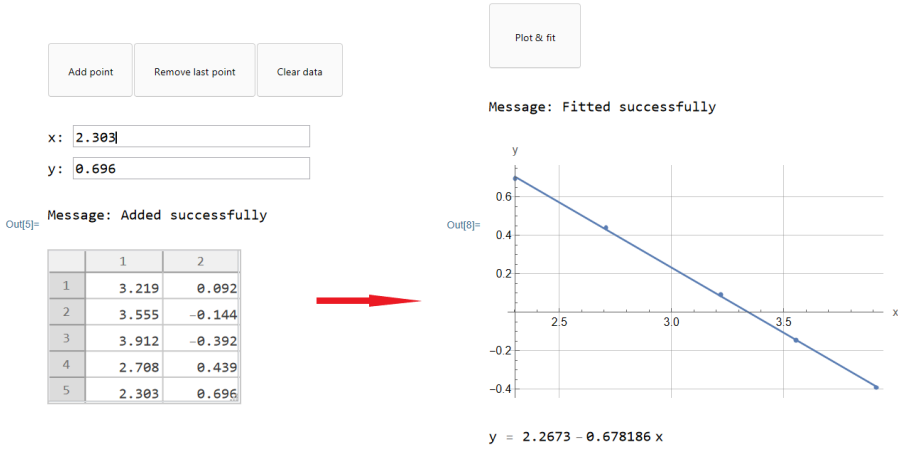
δ	0.100	0.080	0.060	0.120	0.150
k	−1.08567	-1.11796	-1.2865	-0.942494	-0.903676
$\ln\delta$	-2.303	-2.526	-2.813	-2.120	-1.897
$\ln\|k\|$	0.092	0.112	0.252	-0.059	-0.101
Z	0.758	0.826	0.925	0.706	0.647

D_{gb}/D	25	35	50	15	10
k	−1.08567	-0.865926	-0.675888	-1.55123	-2.00586
$\ln[D_{gb}/D]$	3.219	3.555	3.912	2.708	2.303
$\ln\|k\|$	0.092	-0.144	-0.392	0.439	0.696
Z	0.758	0.603	0.473	1.072	1.413

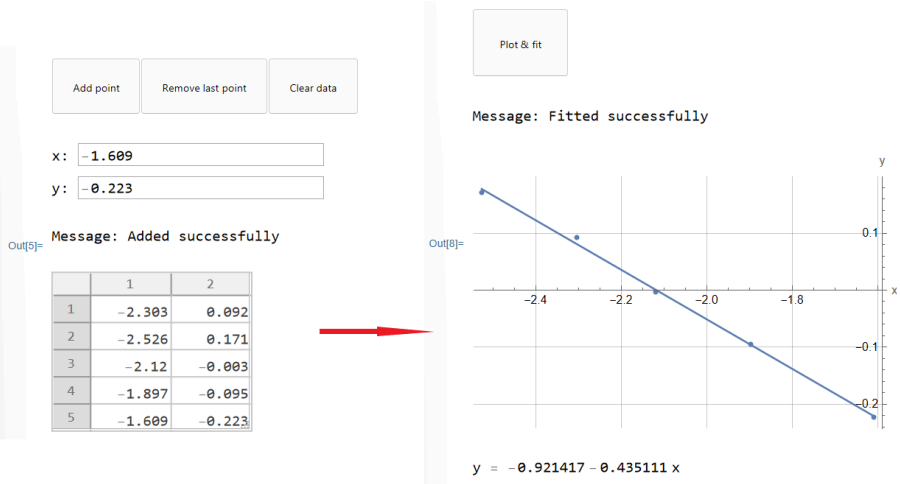
t	0.100	0.080	0.120	0.150	0.200
k	−1.08567	-1.18623	-0.99746	-0.908966	-0.799996
$\ln t$	-2.303	-2.526	-2.120	-1.897	-1.609
$\ln\|k\|$	0.092	0.171	-0.003	-0.095	-0.223
Z	0.758	0.836	0.699	0.634	0.558



$\ln|k|$ against $\ln\delta$.



$\ln|k|$ against $\ln[D_{gb}/D]$.

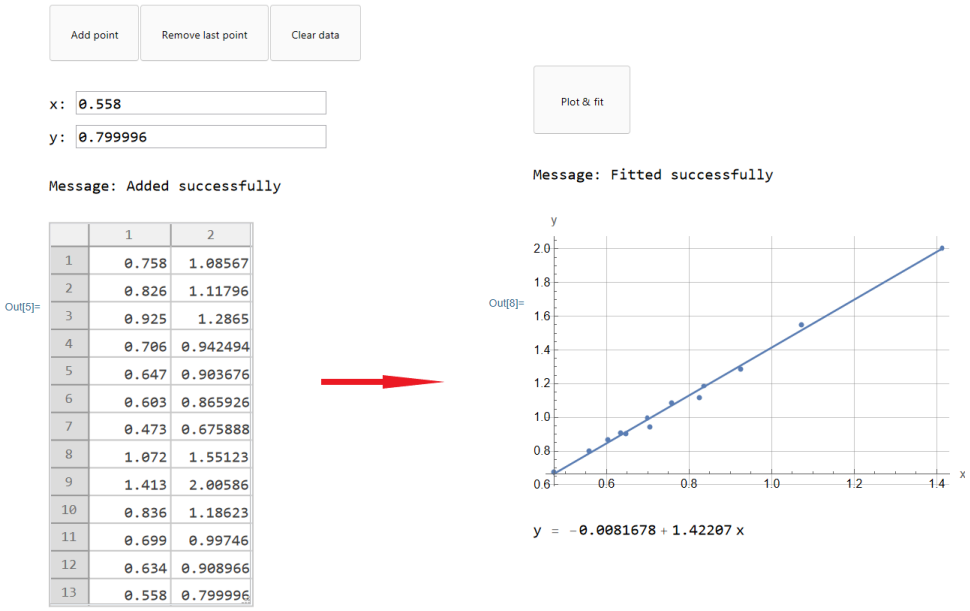


$\ln|k|$ against $\ln t$.

The coefficients we are looking for are the slopes of these lines:

$$\xi = -0.39, \qquad \eta = -0.68, \qquad \zeta = -0.44$$

C3. Knowing ψ, ξ, η and ζ, we find C as the slope of the dependence of $Y = |k|$ on $X = Z = \delta^{-0.39}[D_{gb}/D]^{-0.68}t^{-0.44}$. Calculating X for the points of the previous task, we plot it with `InteractiveBuffer.nb`:



	1	2
1	0.758	1.08567
2	0.826	1.11796
3	0.925	1.2865
4	0.706	0.942494
5	0.647	0.903676
6	0.603	0.865926
7	0.473	0.675888
8	1.072	1.55123
9	1.413	2.00586
10	0.836	1.18623
11	0.699	0.99746
12	0.634	0.908966
13	0.558	0.799996

The fit of $|k|$ against Z.

$$\boxed{C = 1.42}$$

Solution

A1. The energy of a relativistic particle can be expressed in terms of its momentum,

$$E = \sqrt{p^2c^2 + m_\mu^2 c^4},$$

and its velocity equals

$$v = \frac{c^2 p}{E} = \frac{cp}{\sqrt{p^2 + m_\mu^2 c^2}}.$$

For the speed difference we have

$$\boxed{\Delta v = c - v = c\left(1 - \frac{p}{\sqrt{p^2 + m_\mu^2 c^2}}\right) \approx \frac{1}{2} c \left(\frac{m_\mu c}{p}\right)^2 \approx 1.75 \cdot 10^5 \text{ m/s}}$$

For the given numerical values we can use the approximate formula: it deviates from the exact value only in the fourth figure.

A2. We use the expression for the energy from the previous question. Note that for the given numerical values one can use the approximate expression for an ultrarelativistic particle, $E \approx pc$.

$$\boxed{E_k = \sqrt{p^2c^2 + m_\mu^2 c^4} - m_\mu c^2 \approx 2.99 \text{ GeV}}$$

A3. From the relativistic equation of motion

$$\frac{d\vec{p}}{dt} = q\vec{v} \times \vec{B} = \frac{q}{\gamma m_\mu} \vec{p} \times \vec{B}$$

it follows that the absolute values of the velocity and the momentum are constant. Therefore the momentum rotates:

$$\frac{d\vec{p}}{dt} = \vec{\omega}_c \times \vec{p}, \qquad \vec{\omega}_c = -\frac{q\vec{B}}{\gamma m_\mu}.$$

Here ω_c is the angular velocity of the muon. The period is

$$\boxed{T = \frac{2\pi}{\omega_c} = \frac{2\pi m_\mu \gamma}{qB} = 1.49 \cdot 10^{-7} \text{ s}}$$

The orbit radius is

$$\boxed{R = \frac{vT}{2\pi} = \frac{\gamma m_\mu}{qB} v = \frac{\gamma m_\mu}{qB} \cdot \frac{pc^2}{E} = \frac{p}{qB} = 7.12 \text{ m}}$$

A4. Due to relativistic time dilation the muon lifetime in the laboratory reference frame is $t = \gamma\tau$, therefore the number of revolutions is

$$\boxed{N = \frac{t}{T} = \gamma\tau\,\frac{qB}{2\pi m_\mu \gamma} = \frac{qB\tau}{2\pi m_\mu} = 431}$$

A5. The torque acting on a magnetic dipole in a magnetic field is $\vec{M} = \vec{\mu} \times \vec{B}$, thus the equation of motion is

$$\boxed{\frac{d\vec{s}}{dt} = \vec{M} = \vec{\mu} \times \vec{B} = g_\mu \frac{q}{2m_\mu}\,\vec{s} \times \vec{B} = -g_\mu \frac{q\vec{B}}{2m_\mu} \times \vec{s}}$$

A6. From the previous question it follows that the angular velocity is

$$\vec{\omega} = -g_\mu \frac{q\vec{B}}{2m_\mu}.$$

Therefore the z component of the angular velocity is

$$\boxed{\omega_z = -g_\mu \frac{qB}{2m_\mu}}$$

B1. Recall the formula for the addition of velocities from the problem. In our case $v'_x = 0$ (the velocity component of frame C along the x axis), therefore the denominator is $1 + v'_x v_x/c^2 = 1$:

$$\boxed{v_{Cx} = v_x, \qquad v_{Cy} = v'_y\sqrt{1 - v_x^2/c^2} = v'_y/\gamma}$$

B2. Note that if frame B moves relative to A with velocity $\vec{v}$, then frame A moves relative to frame B with velocity $-\vec{v}$. Therefore the velocity of frame B with respect to frame C is $-v'_y$, and the velocity of frame A with respect to frame B is $-v_x$. The γ coefficient is calculated with the velocity v'_y:

$$\gamma = \frac{1}{\sqrt{1 - v_y'^2/c^2}} \approx 1,$$

since we have to keep only the terms linear in v'_y. The velocities of frames A and B are orthogonal and the denominator in the velocity addition formula is equal to 1. The velocity of frame A with respect to frame C is

$$\boxed{v''_{Ax} = -v_x, \qquad v''_{Ay} = -v'_y}$$

B3. The angle between two vectors can be calculated from their vector product (the length of the vector product is $|\vec{a} \times \vec{b}| = |\vec{a}||\vec{b}|\sin\alpha$, where α is the angle between the vectors). Let $\vec{e}_x$ be a unit vector in the x direction, $\vec{e}_y$ in the y direction and $\vec{e}_z$ in the z direction. The vector product is

$$-\vec{v}''_A \times \vec{v}_C = (v_x\vec{e}_x + v'_y\vec{e}_y) \times \left(v_x\vec{e}_x + (v'_y/\gamma)\vec{e}_y\right) = v_x v'_y \left(\frac{1}{\gamma} - 1\right)\vec{e}_z.$$

Both vectors are parallel to the xy plane, therefore the vector product is collinear with the z axis, and a positive z component corresponds to a positive angle of rotation. Up to the terms linear in v_y' we can approximate $|\vec{v}_A''| = |\vec{v}_C| = v_x$, thus the angle of rotation is

$$\boxed{\Delta\theta = \frac{1}{v_x^2}\, v_x v_y' \left(\frac{1}{\gamma} - 1\right) = -\frac{v_y'}{v_x}\left(1 - \frac{1}{\gamma}\right)}$$

B4. After a time dt in the laboratory frame of reference the particle velocity changes by $v_y = a_y\, dt$; in the frame B, which is comoving for the particle at the initial moment of time, the corresponding velocity is $v_y' = \gamma v_y$. This result can be obtained from B1, or directly from the velocity addition formula (the laboratory frame of reference moves relative to frame B with velocity $-v_x$, and the particle velocity with respect to the laboratory frame of reference is $\vec{v} = (v_x, v_y)$):

$$v_y' = \frac{v_y\sqrt{1 - v_x^2/c^2}}{1 - v_x^2/c^2} = \frac{v_y}{\sqrt{1 - v_x^2/c^2}} = \gamma v_y.$$

Thus the rotation angle is

$$d\theta = -\frac{\gamma a_y\, dt}{v_x}\left(1 - \frac{1}{\gamma}\right),$$

and the angular velocity is

$$\boxed{\omega_T = \frac{d\theta}{dt} = -\frac{a_y}{v_x}(\gamma - 1)}$$

Note that according to our sign conventions a positive angle of rotation corresponds to a positive z component of the angular velocity.

B5. From the equation of motion in the magnetic field

$$\frac{d\vec{p}}{dt} = m_\mu \frac{d(\gamma\vec{v})}{dt} = q\vec{v} \times \vec{B}$$

we calculate the acceleration component (note that γ is constant)

$$a_y = -\frac{q v_x B}{\gamma m_\mu},$$

thus the angular velocity is

$$\boxed{\omega_T = \frac{qB}{\gamma m_\mu}(\gamma - 1)}$$

C1. The full angular velocity of precession is the sum of the angular velocity of precession in the magnetic field and the Thomas precession angular velocity:

$$\boxed{\omega_s = \omega_z + \omega_T = -g_\mu \frac{qB}{2m_\mu} + \frac{qB}{\gamma m_\mu}(\gamma - 1)}$$

C2. The muon momentum rotates with an angular velocity equal to the angular velocity ω_c of the muon in the magnetic field, which was calculated in A3; its z component is

$$\omega_{cz} = -\frac{qB}{\gamma m_\mu}.$$

Therefore the angular velocity of the spin with respect to the momentum is equal to the difference between the angular velocities of the spin and the momentum (all rotations are around the z axis):

$$\omega_a = \omega_s - \omega_c = -g_\mu \frac{qB}{2m_\mu} + \frac{qB}{\gamma m_\mu}(\gamma - 1) + \frac{qB}{\gamma m_\mu} = -g_\mu \frac{qB}{2m_\mu} + \frac{qB}{m_\mu}.$$

With the relation $g_\mu = 2(1 + a_\mu)$ we obtain

$$\boxed{\omega_a = -a_\mu \frac{qB}{m_\mu}}$$

C3. The frequency is related to the angular velocity by $f_a = \omega_a / 2\pi$, therefore

$$\boxed{a_\mu = \frac{m_\mu \omega_a}{qB} = \frac{2\pi f_a m_\mu}{qB} = 0.00117}$$

The direction of the spin precession is not given, therefore we cannot determine the sign of a_μ; it turns out that $a_\mu > 0$.

Solution

A1. They are equal to 0 because the field cannot tend to infinity at a large distance from the fiber.

$$\boxed{C_1 = 0, \qquad C_2 = 0}$$

A2. Substituting the form of the solution into the wave equation:

$$\frac{\partial^2 A}{\partial x^2} + \frac{\partial^2 A}{\partial y^2} + \left(\frac{\omega^2 \varepsilon}{c^2} - \beta^2\right) A = 0.$$

Considering that $\partial^2 A/\partial y^2 \ll \partial^2 A/\partial x^2$, and that outside the fiber $\varepsilon = 1$, we get:

$$\boxed{k_{in} = \sqrt{\frac{\omega^2 \varepsilon(\omega)}{c^2} - \beta^2}, \qquad k_{out} = \sqrt{\beta^2 - \frac{\omega^2}{c^2}}}$$

A3. We can write Faraday's law of electromagnetic induction for a rectangular circuit with vertices at the points $(a/2 - \delta a, y, z)$, $(a/2 - \delta a, y, z + \Delta z)$, $(a/2 + \delta a, y, z + \Delta z)$, $(a/2 + \delta a, y, z)$. At $\delta a \to 0$ the derivative of the magnetic flux tends to 0, while the circulation of the electric field strength keeps the same form; hence the circulation is also equal to 0. At $\delta a \to 0$ the circulation can be expressed as $\Gamma = (E_{\tau 1} - E_{\tau 2})\Delta z = 0$, hence

$$\boxed{E_{\tau 1} = E_{\tau 2}}$$

A4.

$$\boxed{B = \exp(k_{out} a/2)\cos(k_{in} a/2)}$$

A5. The proof is similar to A3, but instead of the law of electromagnetic induction, the circulation theorem for the magnetic field strength is used.

A6. Using the differential form of Faraday's law we get:

$$-\mu_0 \frac{\partial H_z}{\partial t} = \frac{\partial E_y}{\partial x}.$$

The derivative of H_z with respect to time can be expressed as

$$\frac{\partial H_z}{\partial t} = i\omega H_z,$$

and therefore

$$\boxed{H_z = E_0 \operatorname{Re}\left(\frac{i}{\mu_0 \omega}\frac{\partial A}{\partial x}\exp(i\omega t - i\beta z)\right)}$$

A7. To find H_z, differentiate A with respect to x:

$$\frac{\partial A}{\partial x} = -k_{in}\sin(k_{in} x), \quad |x| < a/2; \qquad \frac{\partial A}{\partial x} = -k_{out}\exp\big(-k_{out}(x - a/2)\big)\cos(k_{in} a/2), \quad x > a/2.$$

Equating H_z on both sides of the interface:

$$\boxed{\tan(k_{in}a/2) = \frac{k_{out}}{k_{in}}}$$

A8.

$$\boxed{\tan\left(\frac{a}{2}\sqrt{\frac{\omega^2\varepsilon(\omega)}{c^2} - \beta^2}\right) = \sqrt{\frac{\beta^2c^2 - \omega^2}{\omega^2\varepsilon(\omega) - \beta^2c^2}}}$$

B1.

$$\boxed{r_s^2 + t_s^2 = 1}$$

B2.

$$\boxed{B_{in}(t) = \kappa B_{out}(t - \tau(\omega))}$$

B3. Let $\varphi = -\omega\tau$. Then

$$B_{out}(t) = it_sA_{in}(t) - r_sB_{in}(t), \qquad B_{in}(t) = \kappa e^{i\varphi}B_{out}(t),$$

$$\boxed{B_{in}(t) = \kappa e^{-i\omega\tau}B_{out}(t)}$$

B4. Substituting $B_{out}(t)$ from the previous task:

$$B_{in}(t) = \kappa e^{i\varphi}\big(it_sA_{in}(t) - r_sB_{in}(t)\big), \qquad B_{in}(t) = \frac{it_s\kappa e^{i\varphi}A_{in}(t)}{1 + r_s\kappa e^{i\varphi}},$$

$$\boxed{B_{in}(t) = \frac{it_s\kappa e^{-i\omega\tau}A_0e^{i\omega t}}{1 + r_s\kappa e^{-i\omega\tau}}}$$

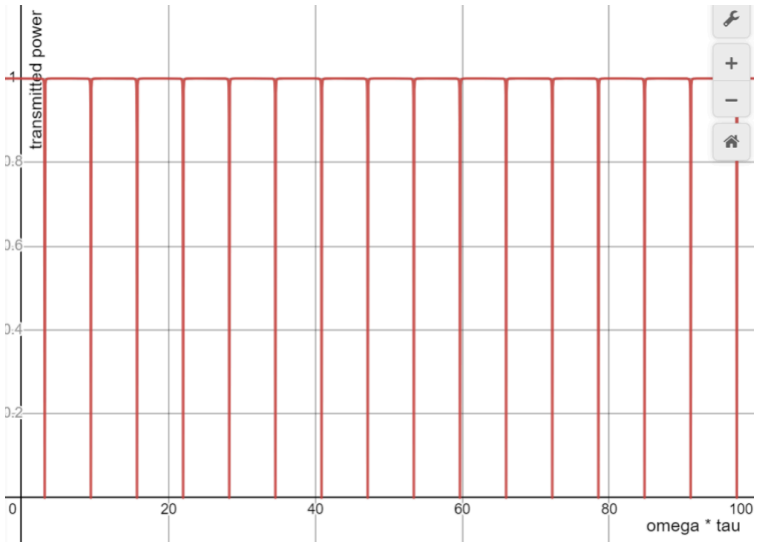
B5. The wave leaving P_2 is

$$A_{out}(t) = -r_sA_{in}(t) + it_sB_{in}(t) = -A_{in}(t)\left(r_s + \frac{t_s^2\kappa e^{i\varphi}}{1 + r_s\kappa e^{i\varphi}}\right).$$

Let us bring the expression in brackets to a common denominator and divide the squared modulus of the numerator by the squared modulus of the denominator:

$$\boxed{\eta = \frac{N_2}{N_1} = \frac{\kappa^2\sin^2(\omega\tau) + \big(r_s + \kappa\cos(\omega\tau)\big)^2}{\kappa^2r_s^2\sin^2(\omega\tau) + \big(1 + r_s\kappa\cos(\omega\tau)\big)^2}}$$

B6.



The transmitted power N_2/N_1 against $\omega\tau$.

$$\boxed{\omega_{res}\tau = \pi(1+2n)}$$

B7. Let us find the depth of the absorption peak by expanding the expression for the power ratio in a Taylor series:

$$\eta(\omega_{res}) \approx \frac{(1-\kappa-t_s^2/2)^2}{(1-\kappa+t_s^2/2)^2} \approx 0, \qquad \Delta\eta_{max} = 1-\eta(\omega_{res}) \approx 1,$$

or by substituting the value of φ from the previous item without expanding: $\eta(\omega_{res}) \approx 1.6\cdot 10^{-6} \approx 0$, $\Delta\eta_{max} \approx 1$.

The sharpness is the ratio of ω to the width of the region in which $\Delta\eta > \Delta\eta_{max}/2$. Let us estimate the width of this region. Put

$$\varphi = \delta + \pi(1+2n), \qquad \delta \ll \pi,$$

and expand in a Maclaurin series in δ:

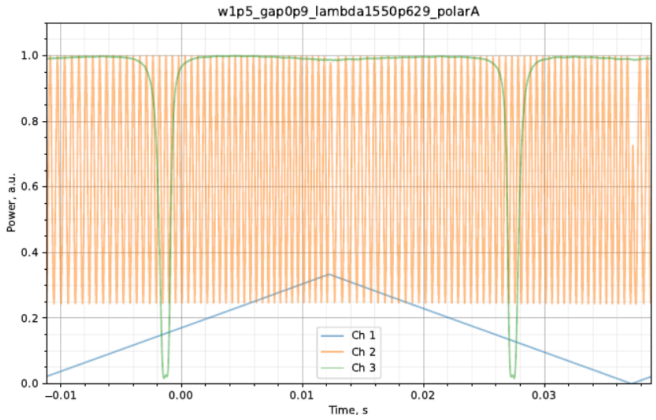
$$\eta \approx \frac{\delta^2 + (1-\kappa-t_s^2/2+\delta^2/2)^2}{\delta^2 + (1-\kappa+t_s^2/2+\delta^2/2)^2} \approx \frac{\delta^2}{\delta^2 + (1-\kappa+t_s^2/2)^2} = \frac{1}{2}.$$

From here we get

$$\delta \approx 1-\kappa+t_s^2/2 \approx 0.01,$$

$$\boxed{Q = \frac{\omega_{res}}{\Delta\omega} = \frac{\pi(1+2n)}{2\delta} \approx \frac{\pi n}{1-\kappa+t_s^2/2} = \pi\cdot 10^4 = 3.14\cdot 10^4}$$

C1.

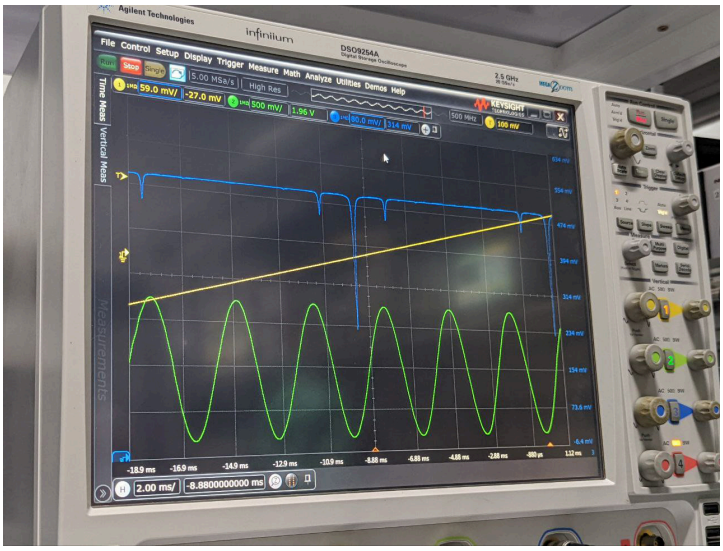


Oscilloscope readings: the interferometer signal and the resonator signal with its absorption dip.

C2. The field amplitude at the exit of the EOM equals

$$E_{EOM}(t) = f(t)\cos(\omega(t)t) = \beta\cos(\omega t) + \frac{1-\beta}{2}\Big(\cos(\Omega+\omega)t + \cos(-\Omega+\omega)t\Big),$$

where $\omega = \omega(t) = \omega_0 + \alpha t$. In this case three waves fall on the FR, with frequencies shifted relative to each other by exactly Ω. Therefore the number of absorption peaks triples: they can be observed at the frequencies ω_{res}, $\omega_{res}+\Omega$, $\omega_{res}-\Omega$, where ω_{res} is the frequency of minimum transmission (defined in B6).



What the oscilloscope shows with the modulator on.

C3. We can measure one period by changing Ω so that the absorption maxima are separated by ω_{MZI} from each other. Then the absolute error is of order $\omega_0/Q = 24$ MHz, and the relative error is

$$\boxed{\frac{\omega_0}{Q\,\Omega_{max}} = 0.02}$$

D1.

$$\boxed{v_g = 1/\beta_1}$$

D2. Let us calculate the derivatives and substitute them into the equation (dividing by $iF_0\exp(i\sigma z)$):

$$\frac{\sigma}{\cosh\theta s}+\frac{\beta_2\theta^2}{2\beta_1^2}\left(-\frac{2}{\cosh^3\theta s}+\frac{1}{\cosh\theta s}\right)=\frac{\gamma F_0^2}{\cosh^3\theta s}.$$

From here we get

$$\boxed{F_0^2=-\frac{\beta_2\theta^2}{\gamma\beta_1^2},\qquad \sigma=-\frac{\beta_2\theta^2}{2\beta_1^2}}$$

D3. Using the result of B3, the natural-frequency condition reads

$$\beta(\omega_\mu)L+\pi=2\pi(\mu+\mu_0).$$

Now let us substitute the expression for $\beta(\omega)$, and into it ω_μ. We get

$$\left(\beta_0+\beta_1\left(D_1\mu+\frac{D_2\mu^2}{2}\right)+\frac{\beta_2 D_1^2\mu^2}{2}\right)L\approx 2\pi(\mu+\mu_0+\text{const}).$$

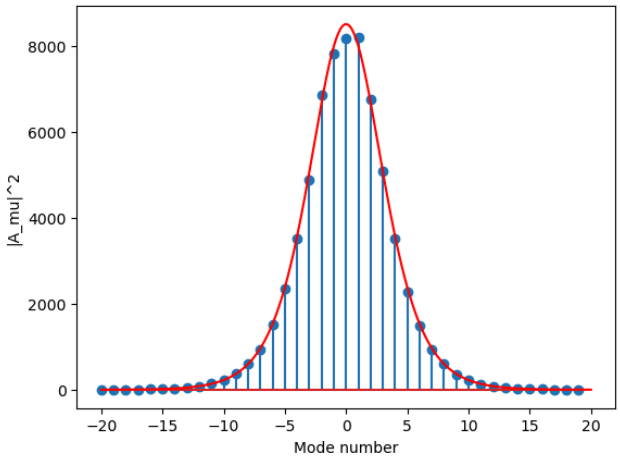
Equating the coefficients of μ and μ^2 on the two sides:

$$\boxed{D_1=\frac{2\pi}{\beta_1 L},\qquad D_2=-\frac{\beta_2 D_1^2}{\beta_1}=-\frac{4\pi^2\beta_2}{\beta_1^3L^2}}$$

D4. $F_0^2>0$, so $\beta_2<0$ (as γ has the sign of $\chi>0$), and then

$$\boxed{D_2>0}$$

D5. The spectrum consists of narrow bands at the natural frequencies of the FR, as shown in the figure. The envelope can be fitted by $F_1/\cosh^2\big(F_2(\omega-\omega_0)\big)$ (here F_1, F_2 are some constants depending on F_0,σ,θ), but an analytical expression for the envelope is not required.



Emission spectrum of the resonator: lines at the natural frequencies $\omega_0+D_1\mu$ (horizontal axis: mode number μ).

D6. We get a ruler with a pitch of D_1, and the absolute error is about half of this pitch:

$$\boxed{\Delta\omega\approx\frac{D_1}{2}}$$

D7.

$$\boxed{\tau_s = \frac{2\pi}{D_1}}$$